PROBLEM SPACE

AI TRIAGE SUPPORT SYSTEM CO-DESIGN CANVAS

Setting A — HCI: Mercer ED Triage Desk

USER

	Group(s)	Characteristics	Needs
primary users	ED triage nurses responsible for assessing patients, recording clinical information and assigning an Emergency Severity Index level.	Clinically trained staff working under significant time pressure. They may assess several patients in rapid succession while recording vital signs, asking questions, managing queues and responding to interruptions. Users may also be affected by fatigue during long or overnight shifts and may have different levels of confidence with digital systems.	• Rapid access to a reliable triage recommendation during patient assessment. • A simple interface that can be understood at a glance without increasing cognitive workload. • Clear explanations of why the AI suggested a particular ESI level. • A quick and visible way to confirm or override the recommendation. • Reliable operation during busy shifts and the ability to continue working if the system becomes unavailable.
secondary users	Emergency physicians, charge nurses, bed managers, Observation Unit staff, clinical IT staff and hospital governance personnel who review triage decisions, alerts or system performance.	These users have different clinical, operational and technical responsibilities. Clinicians require clear and relevant patient information, while IT and governance staff require reliable records, system transparency and evidence that the tool is operating safely.	• Accurate documentation of AI recommendations and final clinical decisions. • An audit trail showing overrides, timestamps and user actions. • Reliable system performance data to support quality improvement and governance. • Confidence that the AI supports clinical decision-making rather than replacing professional judgement.

Goal(s)

	SHORT-TERM	LONG-TERM
primary users	• Complete patient triage efficiently without delaying patient care. • Receive an AI-assisted ESI recommendation that supports rapid clinical decision-making. • Confirm or override the recommendation while maintaining full clinical control.	• Improve consistency and accuracy of triage decisions across different shifts and staff. • Reduce cognitive workload while maintaining patient safety. • Increase confidence in AI-assisted clinical decision support without creating dependence on the system.
secondary users	• Review AI recommendations and nurse overrides for quality assurance. • Monitor system performance and identify workflow issues during pilot deployment.	• Improve consistency and accuracy of triage decisions across different shifts and staff. • Reduce cognitive workload while maintaining patient safety. • Increase confidence in AI-assisted clinical decision support without creating dependence on the system.

AI TRIAGE SUPPORT SYSTEM

Task(s)

SHORT-TERM	LONG-TERM
• Receive patient information from manual entry. • Analyse clinical features using the trained triage model. • Predict the recommended ESI level. • Display confidence level and contributing clinical factors. • Allow the nurse to confirm or override the recommendation.	• Improve consistency of triage decisions. • Reduce cognitive workload during busy shifts. • Maintain a complete audit trail of recommendations and overrides. • Support continuous improvement of the model using reviewed clinical data.

Advantages

What tasks does the AI system perform?

Clinical Support	User Experience	Explainability	Decision Support
• Provides rapid clinical decision support. • Highlights patients requiring urgent review. • Reduces variation in triage recommendations.	• Simple interface that is easy to interpret under pressure. • Reduces cognitive workload during busy shifts.	• Displays confidence score. • Shows the key factors influencing the prediction. • Supports transparent decision-making.	• Produces recommendations within seconds. • Allows immediate nurse confirmation or override. • Logs all clinical decisions.

Data Integration	Workflow Integration	Reliability	Connection to system
• Accepts manually entered patient information. • Can support future EHR integration.	• Fits naturally into the existing triage workflow. • Does not interrupt patient assessment.	• Manual workflow remains available if the system fails. • Low-confidence predictions prompt additional clinical review.	• Can connect to Mercer EHR. • Supports audit logs and reporting. • Stores prediction history for governance purposes.